

Aravind Rajpurohit

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EXPERIENCE

Software Engineer – Undetectable AI

Jul 2024 – Present

- Investigated leads with BigQuery/Splunk and sandboxed PoCs; turned findings into guardrail tests and retraining data, lifting precision/recall by ~8/6 percentage points.
- Implemented distributed pipelines with SQS + Lambda + Redis, boosting throughput 1.6× and lowering timeouts by 28% during peak traffic.
- Collaborated cross-functionally with ML and product teams to ship new API features, improving user engagement and driving measurable retention gains.

Software Engineer – Drexel University (Part time, Full time over summer)

Oct 2022 – Jun 2024

- Developed and maintained internal tools in Python/SQL to clean, validate, and update alumni database records, improving data accuracy by 15%.
- Assisted in system integration and bug fixes, collaborating with senior engineers to ensure smooth migration of alumni data to a new backend system.

Backend Engineer (Capstone) – US Department of Agriculture

Sep 2022 – Jun 2023

- Developed image-processing backend in Python (Scikit-Image, OpenCV) to detect Spotted Lanternfly egg masses from grayscale camera feeds, using connected component analysis (CCA) for blob detection.
- Designed and tested data pipeline to process batches of 100+ images in real time, improving accuracy under varying light/angle conditions and reducing false negatives by ~20%.
- Collaborated with hardware and USDA stakeholders to integrate camera systems with algorithm outputs, ensuring scalable deployment on railway inspections for invasive species control.

SKILLS

- **Backend Development:** Python, Node.js, TypeScript, Java, SQL, REST APIs, gRPC, OAuth2/OIDC, RBAC, rate limiting, token revocation, structured logging, evidence auditing.
- **Cloud & Infrastructure:** AWS (Lambda, S3, CloudWatch, IAM, KMS, Secrets Manager), GCP (Pub/Sub, Cloud Functions, BigQuery, Cloud Logging), Docker, Kubernetes, Redis, PostgreSQL, WAF/Cloud Armor, VPC-SC.
- **Systems & Data:** Distributed systems, event-driven pipelines, Elasticsearch, Kafka, SIEM (Splunk, Chronicle), telemetry pipelines, anomaly detection, Sigma, YARA, OpenCV, Scikit-Image.
- **Collaboration & Practices:** Agile/Scrum, CI/CD (GitHub Actions, Terraform), Secure SDLC, SAST/DAST, Git, Cross-Functional Teamwork.
- **AI & Machine Learning:** Pandas, NumPy, scikit-learn, TensorFlow, Keras, HuggingFace, LangChain, prompt-injection/jailbreak defenses, PII redaction, output filtering, safety evaluation harnesses.
- **Certifications:** CompTIA Security+ (Credential ID: COMP001022677772)

RECENT PROJECTS

[GitHub](#)

Container Orchestration Engine

(Docker SWARM . AWS . Flask . MongoDB . Kubernetes)

- Designed a Kubernetes-like container orchestrator with load balancing, fault tolerance, and auto-scaling, and deployed a microservices-based photo-sharing platform using Docker Swarm and REST APIs.

Meteorological Data Analysis

(Python . Pandas . NumPy . Matplotlib . Seaborn . Jupyter)

- Conducted end-to-end analysis of climate datasets by cleaning and preprocessing raw data, applying statistical methods to detect seasonal trends and anomalies, and visualizing results through dashboards to support data-driven insights.

EDUCATION

M.S. Cybersecurity | B.S. Information Science & Engineering

Drexel University, Philadelphia, PA

Coursework: Software Engineering, Artificial Intelligence, Machine Learning, Cybersecurity Forensics, Data Structures and Algorithms, Web Development, System Fundamentals, Calculus I–II, Linear Algebra.